



Potential of zeotropic mixtures as working fluids in organic Rankine cycles

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ABSTRACT

The effect of using mixtures as working fluids in organic Rankine cycles (ORCs) is examined with a simulation model of the cycle that includes all elements of an actual installation. We consider several of the commonly used pure ORC fluids as potential components, discuss a mixture selection method, and suggest optimal concentrations. For heat sources at 150 °C and 250 °C, a potential increase of 16% and 6% in cycle efficiency is found. The electricity production at optimal thermal power recuperation can be increased by 20% for the low temperature heat source.

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1. Introduction

In the last several years, in a context of rising electricity demand and cost, the generation of power from industrial waste heat and geothermal or solar sources has attracted an increasing amount of attention. The organic Rankine cycle (ORC) [1] is a viable answer, when dealing with heat sources at low temperature levels, i.e., below 300 °C.

The total efficiency of a Rankine cycle [1] is a function of the temperatures of the heat source and sink, the performances of the cycle components, and the thermophysical properties of the working fluid. Until recently, mostly pure fluids, i.e., consisting of a single chemical substance, were considered as the operating medium of ORC installations [2–21].

Possibly the largest shortcoming of a pure fluid in its application in an ORC, is the fact that the evaporation and condensation processes occur isothermally. As a result, the tilted temperature profiles of the heat source and sink cannot be approached closely by the temperature profiles of the fluid in the evaporator and condenser stages, leading to large irreversibilities. The use of mixtures of properly chosen components as the ORC medium can partially resolve this problem.

Azeotropic mixtures, e.g., Solkatherm (pentafluorobutane/perfluoropolyether), are already being used and proven in actual installations [22]. However, these still have isothermal phase transitions.

Zeotropic mixtures, on the other hand, are characterized by non-isothermal phase transitions at constant pressure. These fluids are excellent candidates to match the heat source and sink profiles and hence reduce the irreversibilities, promising a better cycle performance. The potential of mixtures – and their non-isothermal phase transitions – is already exploited in cryogenic refrigeration, where they are at the basis of improved cooling performance [23].

Only a few studies on the use of mixtures as ORC working fluids have been published. In the domain of hydrocarbon mixtures, Li et al. [24] compared the cycle efficiency of R141b/RC318 mixtures with that of three pure fluids. It was found that the use of mixtures allows a wider selection of working fluids. The increase in thermal and exergy efficiency by adding a recuperator was higher for the mixture R141b/RC318 than for R141b. For a low temperature solar ORC, Wang et al. [25] compared three different mixture compositions of dry R245fa and wet R152a to pure R245fa, in a range from 25 °C (condensation temperature) to 85 °C (vaporization temperature). The main benefits of the mixtures were that the cost of the cycle could be reduced as smaller expanders are suited and that the range of usable fluids increased. In an experimental study of a low temperature ORC by the same authors [26], solar collector efficiencies and overall efficiencies of zeotropic R245fa/R152a mixtures were found to be higher than for pure R245fa. An increase of up to 29.1% in energy production by using the mixtures was observed. Heberle et al. [27] performed detailed simulations for

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isobutane/isopentane and R227ea/R245fa mixtures to investigate the exergy efficiency. It was shown that for a heat source temperature below 120 °C, the efficiency increase can reach 15%. Mixtures of either hydrocarbons or siloxanes were investigated by Angelino and Colonna [28]. For multicomponent mixtures, enhanced performance was observed, as the cooling device required less power.

Still, the research into zeotropic mixtures is limited and mainly concerned with specific cases. In this paper, we present a systematic and comparative study of the potential of zeotropic mixtures as working fluids in ORCs, considering several of the commonly used hydrocarbon and siloxane substances as components in various concentrations. We investigate the impact on the operation and efficiency of an ORC, using a purposely designed tool that takes into account all elements of an actual cycle. Two types of heat source are considered, at different temperature levels. We include an optimization of the recuperated heat, which yields upper bounds for the electricity production.

2. Fluid characteristics

A crucial parameter in the performance of an ORC is the working fluid. The medium not only needs to possess the right thermophysical properties (e.g. dry or wet fluid, molar mass, viscosity, etc.), it also has to be chemically stable at the desired working temperature. These requirements have already been discussed extensively in studies of pure fluids, ref. [9–21], and will not be repeated here. To guarantee a chemically and operationally stable ORC medium, each separate component of a potential mixture needs to meet the same requirements. Moreover, as the features and complexity of a mixture and its constituents are more or less similar, we have chosen to include only components in our study that behave well as a pure medium. To ensure mixability, the selected components are either hydrocarbons or siloxanes. See Table 1 for a list of considered pure fluids and their thermophysical properties. Due to the dry nature of the proposed fluids (characterized by a positive slope of the saturated liquid and vapor line in the temperature–entropy diagram), we assume no superheating is required during the ORC operation. Mago et al. [4] indicated that superheating dry fluids leads to a decrease in efficiency.

As already mentioned in the introduction, the main advantage of mixtures as ORC working fluids stems from their non-isothermal phase transitions during vaporization and condensation. This effect becomes apparent when comparing the temperature–entropy diagrams of the cycle processes, for example for pure pentane and for the mixture pentane/hexane (0.5/0.5), see Fig. 1. It is clear that the temperature profile of the mixture [Fig. 1 (a)] follows the heat source profile in a better way than that of the pure fluid [Fig. 1

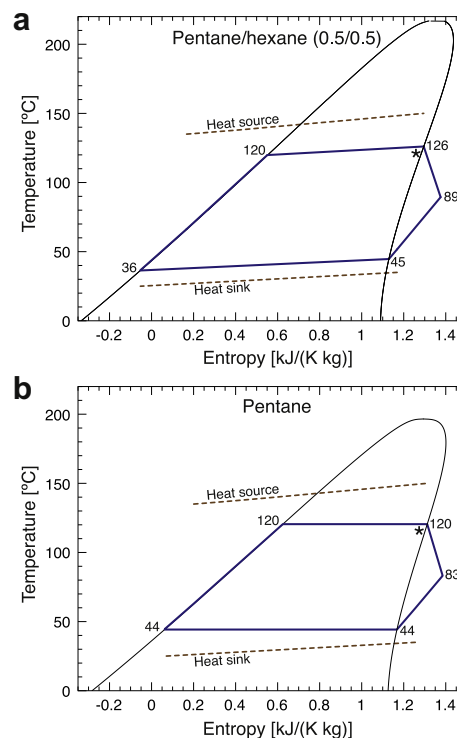


Fig. 1. Temperature–entropy diagrams of the ORC process for (a) the mixture pentane/hexane (0.5/0.5) and for (b) pure pentane as working fluid. The heat source and sink profiles (dashed curves) are included in the diagram.

(b)]. This leads to less exergy losses and, as apparent in Fig. 1, to higher turbine inlet temperatures (Fig. 1, marked by *) due to the smaller temperature difference across the heat exchanger. This in turn results in a higher total efficiency. Similarly, at the condensation side, the non-isothermal behavior of the mixture results in heat rejection at a lower (average) temperature, again increasing the total efficiency.

For cryogenic refrigeration, methods have already been established for some time [23,30] to specify suitable components for mixtures as cooling fluids. We have adopted these guidelines in our assessment of ORC mixtures. One requirement is the appearance of a volatile component (the first component) at 1–1.5 bar, so that low temperatures after expansion can be obtained without the need to reach vacuum pressures. For the second component, the boiling point needs to exceed both the desired average condensation temperature and the boiling point of the first component. By adding a component with a higher boiling point, more heat can be absorbed per mass unit of fluid, hence the flow rate of the ORC medium can be lower [23]. We will also investigate adding a third component, but best results are mostly achieved in small amounts, “tweaking” the phase transition gradients to a desired angle.

3. Cycle

The ORC installation that is considered in this study is schematically represented in Fig. 2. The pump (Fig. 2, no. 1) pressurizes the ORC medium, which is then lead through the internal heat exchanger (no. 2, also known as recuperator), where it can absorb heat that is recovered from the stage after expansion. Next, in the evaporator (no. 3), the heat source brings the fluid to its evaporation point, turning it into vapor. Operational pressures and related fluid temperatures depend on the temperature of the heat source, specified at in and outlet of the heat exchanger. Saturated vapor then

Table 1

Thermophysical properties of the selected pure fluids [29]. Listed are the molar mass, critical temperature and pressure, and the boiling point at atmospheric conditions.

Fluid	<i>M</i> [g/mol]	<i>T</i> _{crit} [°C]	<i>p</i> _{crit} [bar]	<i>T</i> _{evap} @ 1 bar [°C]
Cyclohexane	84.2	280.5	40.8	80.3
Cyclopentane	70.1	238.5	45.2	48.8
Hexane	86.2	234.7	30.3	68.3
HMDS	162.4	245.5	19.4	99.8
Isobutane	58.1	134.7	36.3	−12.1
Isohexane	86.2	224.6	30.4	59.8
Isopentane	72.1	187.2	33.8	27.4
OMTS	236.5	290.9	14.2	152.0
Pentane	72.1	196.6	33.7	35.7
R245fa	134.0	154.0	36.5	14.8
R365mfc	148.1	186.9	32.7	39.8
Toluene	92.1	318.6	41.3	110.1

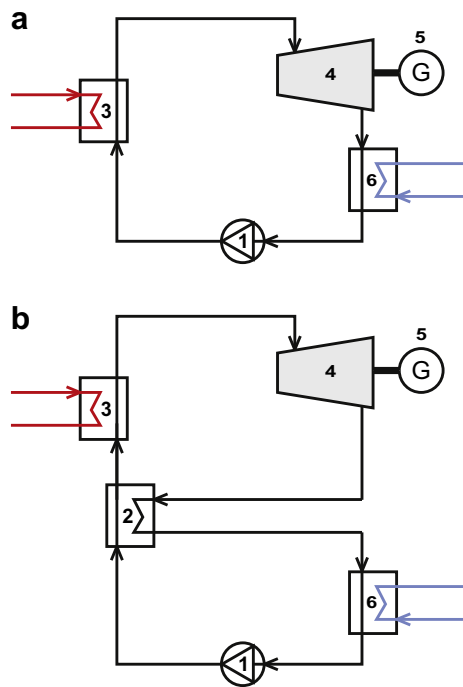


Fig. 2. Cycle diagram of the ORC, (a) without a recuperator, (b) with a recuperator.

enters the turbine (no. 4) and expands there to a superheated vapor, while delivering work that is used to generate electricity (no. 5). Finally, after being passed through the recuperator, the medium is condensed to the liquid state in the condenser (no. 6) using a heat sink. The included recuperator (no. 2) is a regularly used component to increase the performance of an ORC. The effectivity of the recuperator is specified by the temperature difference ΔT_{recup} between the fluid entering the condenser and the fluid coming from the pump.

To analyze the influence of the heat source temperature level on the applicability and potential of different fluid mixtures, we modeled two heat source types. The first is a low temperature heat source. The heat source medium (in this case, water) enters the counterflow type heat exchanger at a temperature of 150 °C and exits the heat exchanger at 135 °C, as heat is transferred from this medium to the ORC fluid for its evaporation. In general, the amount of transferred heat depends on the type and mass flow of the heat source medium, the ORC medium, and the heat exchanger design, and is fixed here at 966 kW. We assume that both the condenser and the evaporator are in counterflow, and possess a finite heat capacity flow rate, in order to achieve a positive effect on the ORC performance by using zeotropic mixtures as the working fluid. The corresponding cycle conditions are given in Table 2. The second

type of heat source has a higher temperature and provides 1287 kW of thermal power. The temperature gradient over the heat exchanger is from 250 °C to 180 °C. Its cycle is described by the parameters in Table 3.

We have chosen heat source and sink types with fixed and equal temperature drops to compare fluids on a purely thermodynamic basis. Under these conditions, the ORC will not optimally recover heat from the source. Later in this paper, the heat source temperatures will be varied to investigate their influence on the applicability and potential of predefined mixtures in more detail.

We have chosen to include a recuperator in the cycles with higher heat source outlet temperature. In the second part of this study, when the heat source outlet temperature is decreased to optimally recover heat, we compare results with and without a recuperator. Using a recuperator can make sense in circumstances where the heat source cannot be cooled down further or when this is undesirable. This is the case when the heat source is in the form of, e.g., engine cooling or thermal solar collectors, or when a part of the heat can be utilized for other applications. For flue gas sources, corrosion issues can pose a limit on the temperature.

Note that the same component efficiencies are chosen for the two heat source types, to enable an objective comparison. In reality, an ORC applied to a higher temperature heat source often has a turbine with a dedicated design, with isentropic efficiencies of 75% or more.

4. Simulation and assumptions

We developed a tool to simulate the described cycle. The FluidProp library [31], in combination with the REFPROP 8.0 database [29], provided the thermophysical data of the considered mixtures and pure fluids. Central in our calculations is a pinch analysis [32] of the heat flows. A nonlinear optimization algorithm enables the determination of the maximal net power output and corresponding operating pressures with respect to the preset pinch temperatures. The same algorithm can be used to find the mixture composition that maximizes the cycle efficiency. We define the cycle efficiency as the ratio of the net output power (electricity produced by the generator at the turbine minus the consumed electricity by the pump) over the thermal input power (source heat transferred to the ORC medium in the evaporator). Furthermore, the tool offers the possibility to analyze the temperature profiles in the evaporator, condenser and recuperator. Appendix A provides details of the model's equations. An error analysis is given in Appendix B.

The performance of the cycle is evaluated assuming steady state conditions of all components, described by constant values for the parameters (see Table 2 and Table 3). We also assume that no pressure or heat losses occur between cycle components and that mass and energy are conserved in each cycle component. The temperatures and pressures of the ORC medium are determined by

Table 2

Parameters for the low temperature heat source, corresponding heat sink and applied ORC. The heat source is water at 5 bar. The heat sink is water at 4 bar.

Heat source	Inlet temperature	150 °C
	Outlet temperature	135 °C
	Mass flow	15 kg/s
Heat sink	Inlet temperature	25 °C
	Outlet temperature	35 °C
Cycle	Isentropic pump efficiency	80%
	Isentropic turbine efficiency	65%
	Generator efficiency	97%
	Pinch evaporator	20 °C
	Pinch condenser	10 °C
	ΔT_{recup}	15 °C

Table 3

Parameters for the high temperature heat source, corresponding heat sink and ORC. The heat source is air at 1 bar. The heat sink is water at 5 bar.

Heat source	Inlet temperature	250 °C
	Outlet temperature	180 °C
	Mass flow	25 kg/s
Heat sink	Inlet temperature	35 °C
	Outlet temperature	50 °C
Cycle	Isentropic pump efficiency	80%
	Isentropic turbine efficiency	65%
	Generator efficiency	97%
	Pinch evaporator	30 °C
	Pinch condenser	20 °C
	ΔT_{recup}	20 °C

the heat source and sink temperature profiles, offset by preset pinch temperatures, which are defined as the minimum temperature differences over the heat exchangers. Our study focuses on the overall cycle performance, rather than on the optimization of individual cycle components (i.e. the heat exchangers, turbine and pump). This pinch methodology ensures an objective thermodynamic comparison, based solely on the differences in fluid properties.

5. Results and discussion

Before we report and discuss the results for all considered mixtures, the selection method is explained via a few specific cases. At the end of this section, we extend the heat source types (Table 2 and Table 3) with varying temperature gradients and examine the optimization of the electricity production.

5.1. Selection of mixture composition

All examples presented here to explain the selection method are under the conditions of the low temperature heat source type (Table 2). We begin with the analysis of the mixture pentane/hexane as the ORC medium. In Fig. 3(a), the dependence of the cycle efficiency (bullets and solid curves) and of the operational pressures (triangles and dotted curves) on the component concentration of the mixture is illustrated. Concentrations will be expressed in mole fractions throughout this paper. Reported efficiencies are under optimal cycle conditions, with evaporation and condensation pressures corresponding to maximum net power output. In this part of the paper, we report the performance in terms of efficiency, for reasons of familiarity. For fixed values of recovered heat, maximum efficiency corresponds with maximum net power output. At the complete left (right) of the curves, the working fluid is pure hexane (pentane). A higher (lower) concentration of the

more volatile component (pentane in this case) gives rise to higher (lower) condensation and evaporation pressures. From an operational point of view, however, extreme pressures are undesirable. Therefore, an upper limit of 90% of the critical pressure of the fluid is imposed in the assessment of potential mixtures. For the pentane/hexane mixture under the specified conditions, all the pressure ratios are acceptable, and the mixture that yields the highest efficiency can be retained.

In Fig. 3 (b), we show the relation between the cycle efficiency and the non-isothermal phase transition gradient or “glide” during the evaporation and condensation stages of the process (circles and dashed lines) for the pentane/hexane mixture. This glide is defined as the temperature increase (or decrease) of the working fluid during the phase transition step. For this particular case, the efficiency maximum coincides with the maxima of the temperature gradients during the phase transition stages and appears for approximately equal fractions of the components. The corresponding mixture composition delivers the best match between these temperature gradients and the temperature profiles of the heat source and sink. In general, the maxima in the phase gradients do not necessarily occur for equal fractions of the components. The example of a R245fa/pentane mixture illustrates this [see Fig. 3 (c)]: here, a ratio of 0.24/0.76 produces maximal phase transition gradients. Furthermore, although often the case, maximum gradients are, in general, not equivalent with an optimal match with the temperature profiles and hence maximum efficiency.

The glide of a mixture is larger when the evaporation temperatures of its components are further apart. In practice, however, this difference between boiling points is limited, as is made apparent by the case of an isopentane/cyclohexane mixture. The boiling point difference is 53 °C (compare with 33 °C for pentane/hexane). As is clear from Fig. 3 (d), the mixture with maximum efficiency does not have the highest phase transition slopes. We note that for this case, the recuperator pinch was set to 20 °C instead of 15 °C (other

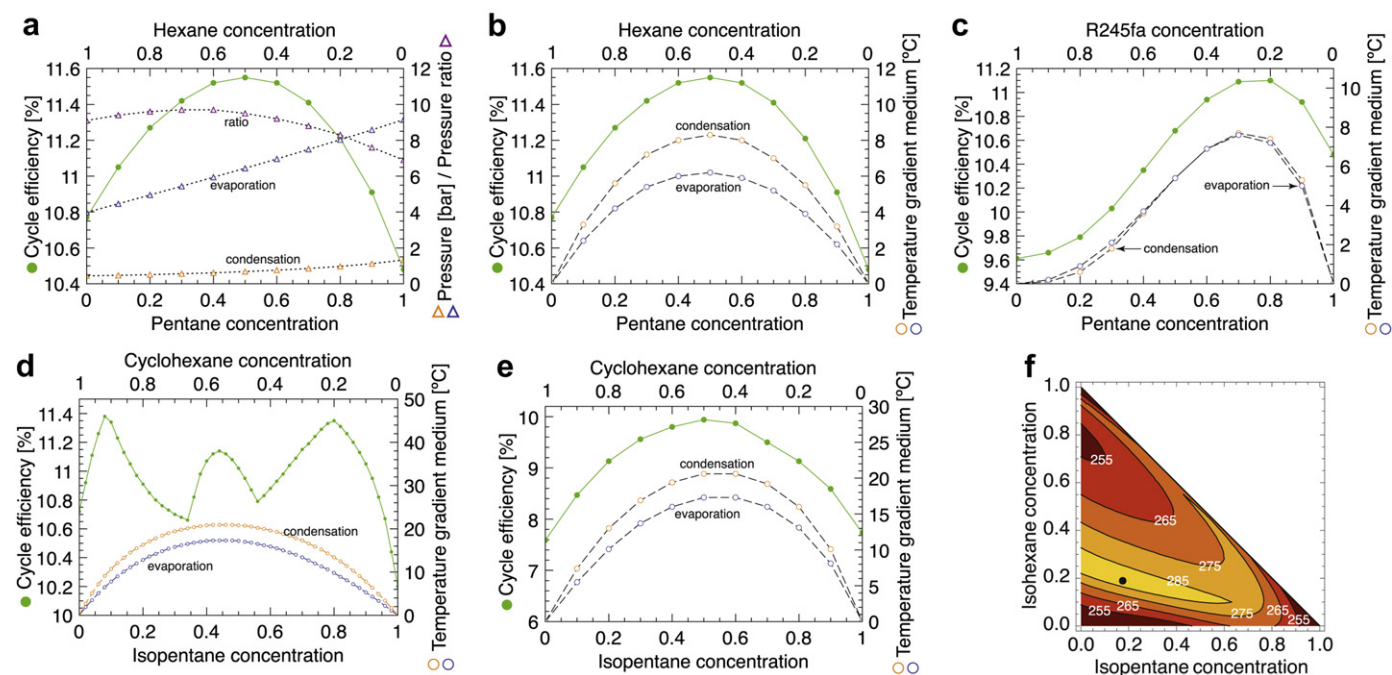


Fig. 3. (a–e) Mixture composition (horizontal axis) versus cycle efficiency (bullets and solid lines, left vertical axis). Included are (a) evaporation pressure, condensation pressure, and pressure ratios (triangles, dotted lines, right vertical axis) (b–e) temperature gradient over evaporation and condensation stage (circles, dashed lines, right vertical axis). (f) For R245fa/isopentane/isohexane, the net electricity production [kW] versus the isopentane and isohexane concentrations; the dot represents the optimal mixture composition, curves are at constant net electricity production. Parameters for all examples as in Table 2 (low temperature heat source), except for (e), where a lower heat source outlet temperature of 125 °C and a higher heat sink outlet temperature of 50 °C are applied.

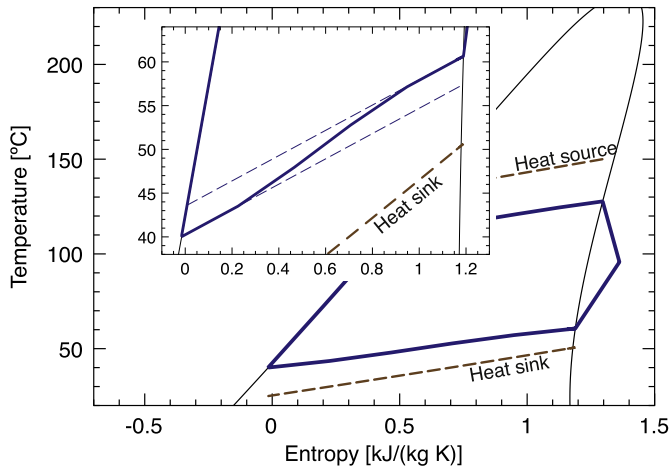


Fig. 4. Fractionating of isopentane/cyclohexane in the condenser.

parameters as in Table 2) to avoid condensation in the recuperator, which, although useful in practice, would bias our thermodynamic comparison of fluids. For this case, the glides for some mixture compositions can become too large relative to the slopes of the heat source and sink temperature profiles. In those circumstances, the smallest attainable temperature difference between the ORC medium and the heat source (or sink) forms a limiting factor and requires a reduction (increase) of the evaporation (condensation) pressure. The irregularities in the graph have their origin in a fundamental shift in the pinch point location.

Mixtures of components with large differences in evaporation temperatures may still be appropriate for heat sources with steep temperature profiles. Caution is warranted, however, as undesirable “fractionating” (where one component is largely in the vapor

state, while the other is still mostly liquid) may occur in the evaporator or condenser stages. A leakage in the system would then result in a rapid composition shift of the ORC medium. Therefore, mixtures that fractionate substantially, are excluded. To illustrate the above, we plotted the cycle efficiency for isopentane/cyclohexane in Fig. 3 (e), this time for a heat source with inlet and outlet temperatures of 150 °C and 125 °C and heat sink inlet and outlet temperatures of 25 °C and 50 °C (Recuperator pinch at 20 °C). Although again a correspondence between the glide curves and the efficiency curve is found, a close look at the process reveals that sudden fractionating occurs: see the temperature–entropy diagram of Fig. 4 for a detailed view of the condensing process.

For 3-component mixtures, two components have to be considered in the optimization (the third is then also determined). In Fig. 3 (f), we show the maximally achievable electricity production (instead of the cycle efficiency for practical reasons) as a function of the concentrations of isopentane and isohexane for a R245fa/isopentane/isohexane mixture. The profiles in higher dimensions are often convoluted and care is taken to retrieve the global maximum.

In what follows, mixtures with vaporization and condensation gradients lower than the heat source and sink gradients, respectively, are retained. In addition, a maximum boiling point difference of 45 °C between components is imposed to avoid significant mixture fractionating.

5.2. Mixture comparison

Here, we present a comparison of the performance of the ORC for several fluids over fixed and identical temperature intervals over the heat sink and source. The objective is not the optimization of the working conditions of an ORC for (waste) heat recovery. We discuss optimal heat recovery at the end of this section, which entails cooling down the heat source further. For this part of the

Table 4

Simulation results for the ORC applied to the low temperature heat source. Listed are the ORC medium composition, optimal concentrations, and corresponding evaporation pressure, condensation pressure, pressure ratio, medium mass flow rate, pump power, generator power, and cycle efficiency.

Medium	C [mole frac.]	P_{evap} [bar]	P_{cond} [bar]	P_{ratio}	$\dot{m}$ [kg/s]	P_{pump} [kW]	P_{gen} [kW]	η_{cycle} [%]
Isobutane	1	30.8	5.9	5.2	2.59	−15.3	98.1	8.58
R245fa	1	20.3	2.9	7.1	4.47	−7.5	100.3	9.61
Isopentane	1	11.0	1.7	6.4	2.40	−4.7	105.0	10.38
R365mfc	1	9.4	1.2	8.1	4.25	−3.6	104.1	10.40
Pentane	1	9.2	1.3	6.9	2.27	−3.7	104.9	10.48
Cyclopentane	1	6.5	0.9	7.5	2.08	−2.0	104.7	10.63
Isohexane	1	4.9	0.6	8.3	2.40	−2.0	105.6	10.73
Hexane	1	4.0	0.4	9.1	2.26	−1.6	105.6	10.77
Cyclohexane	1	2.8	0.3	9.6	2.13	−1.6	105.7	10.85
Toluene ^a	1	1.3	0.1	13.1	2.04	−0.9	105.2	10.86
Solkatherm ^b	1	9.7	1.3	7.3	5.70	−0.4	106.4	10.58
R245fa-R365mfc	0.30–0.70	13.1	1.5	8.7	4.28	−5.0	109.5	10.82
R245fa-isopentane	0.20–0.80	14.7	2.0	7.2	2.83	−6.3	110.8	10.82
Isobutane-isopentane	0.26–0.74	15.3	2.2	7.0	2.38	−6.6	112.8	10.99
R245fa-pentane	0.24–0.76	13.3	1.7	8.0	2.78	−5.4	112.7	11.12
Isopentane-isohexane	0.43–0.57	7.6	0.9	8.8	2.35	−3.2	114.2	11.49
Pentane-hexane	0.50–0.50	6.4	0.7	9.5	2.22	−2.6	114.1	11.55
Isopentane-cyclohexane	0.79–0.21	9.1	1.1	8.2	2.29	−3.6	115.3	11.57
Isopentane-hexane	0.61–0.39	8.1	0.9	8.8	2.29	−3.3	115.1	11.58
R245fa-isopentane-isohexane	0.05–0.64–0.31	10.0	1.2	8.4	2.45	−4.2	115.5	11.52
R245fa-pentane-hexane	0.01–0.52–0.47	6.8	0.7	9.7	2.23	−2.7	115.4	11.67
Isopentane-isohexane-cyclohexane	0.11–0.44–0.45	4.7	0.4	10.5	2.22	−1.7	115.1	11.74

^a Toluene has very low subatmospheric condensation pressure under the considered conditions and is only included for comparison.

^b Solkatherm is an azeotropic mixture.

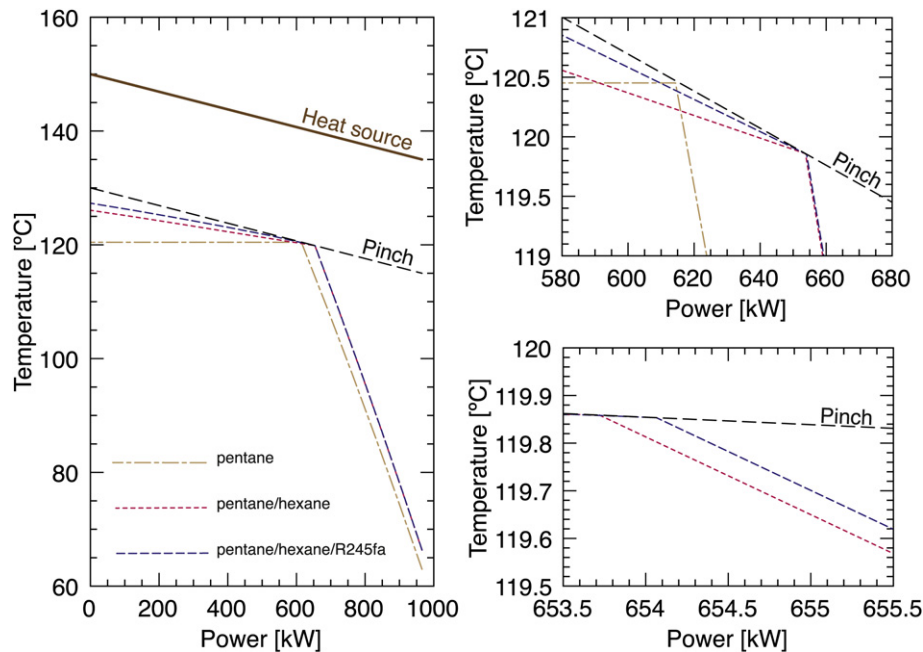


Fig. 5. Power-temperature diagrams of pentane, pentane/hexane and pentane/hexane/R245fa during heating and vaporization of the ORC medium. The pinch lines indicate minimal temperature differences between the heat source and the ORC medium.

study we have included a recuperator in the cycle, as the outlet temperature of the heat source is still rather high.

The results for the examined fluids, suitable for a lower temperature heat source ($T_{in} = 150\text{ °C}$, $T_{out} = 135\text{ °C}$), are presented in Table 4. For comparison, several single component fluids are included. In the second column of Table 4, the mole fraction concentrations that correspond with the highest cycle efficiencies, according to the procedure described in Section 5.1, are listed. Optimal binary mixtures show a 5.2–12.3% increase in generator power and a 4.0–15.7% increase in cycle efficiency with respect to their pure constituents. Compared with the binary mixtures, three-component mixtures only yield a small additional increase in generator power and efficiency.

The mixture isopentane/isohexane (0.43/0.57) gives a cycle efficiency of 11.49% (compared with 10.38% for isopentane, representing a relative performance increase of 10.7%). The generated electricity for the mixture (114.2 kW) is also substantially higher than for pure isopentane (105.0 kW, an increase of 8.8%). Adding a third component to obtain the mixture isopentane/isohexane/cyclohexane, results in an efficiency of 11.74% and an electricity production of 115.1 kW (respectively an additional increase of 2.2% and 0.8% compared to the two-component mixture). Considering now another example, when hexane is added to pure pentane to create the mixture pentane/hexane (0.5/0.5), an efficiency increase of 10.2% and an increase in generated electricity of 8.8% (up to 114.1 kW in comparison with 104.9 kW for pure pentane) is observed. With the addition of R245fa as a third component, an increase of efficiency and generated electricity of only 1.0% and 1.1%, respectively, can be noticed. Note that also in comparison with the azeotropic mixture Solkatherm, a substantial increase in the ORC performance can be achieved by using zeotropic mixtures.

The beneficial effect of using a mixture instead of pure fluids becomes apparent in Fig. 5, where the corresponding power–temperature diagrams of pentane, pentane/hexane and pentane/hexane/R245fa are reproduced. The temperature gradients along the phase transition steps for the mixture pentane/hexane (0.5/0.5) follow the heat source and sink profiles much better than the isothermal steps

of pure pentane. This leads to fewer exergy losses. Adding R245fa to the pentane/hexane mixture results in an even better match of the temperature profiles, however, the improvement is small.

For the higher temperature heat source ($T_{in} = 250\text{ °C}$, $T_{out} = 180\text{ °C}$), with corresponding cycle parameters as in Table 3, our simulation results for compatible single fluids and mixtures are listed in Table 5. Also here, we observe an increase in the generator power and cycle efficiency by using mixtures instead of pure fluids.

For instance, the specified ORC with the pure siloxane fluid hexamethyldisiloxane (HMDS) as its medium, can generate 249.3 kW of electricity with a cycle efficiency of 13.4%. When another siloxane component, octamethyltrisiloxane (OMTS), is added to create a mixture 0.7/0.3 HMDS/OMTS, an electricity production of 260.3 kW and a cycle efficiency of 14.2% are found, an increase of 4.4% and 5.4%, respectively.

Similarly, if the hydrocarbon mixture toluene/cyclohexane (0.6/0.4) is considered for a cycle with parameters as in Table 3, 253.4 kW of electricity can be produced with an efficiency of 13.9%. Compared with the same cycle using pure toluene (generating 239.1 kW with 13.2% efficiency), this represents a power production increase of 6.0% and an efficiency increase of 5.5%.

Table 5

Simulation results for the ORC applied to the high temperature heat source. Listed are the ORC medium composition, optimal concentrations, and corresponding evaporation pressure, condensation pressure, pressure ratio, medium mass flow rate, pump power, generator power, and cycle efficiency.

Medium	C [mole frac.]	p_{evap} [bar]	p_{cond} [bar]	p_{ratio}	$\dot{m}$ [kg/s]	P_{pump} [kW]	P_{gen} [kW]	η_{cycle} [%]
Cyclopentane	1	21.2	1.85	11.5	3.84	−13.3	247.5	13.02
Toluene	1	4.6	0.26	17.6	3.76	−2.5	239.1	13.15
Cyclohexane	1	9.5	0.70	13.6	3.92	−5.9	244.7	13.28
OMTS	1	2.1	0.05	38.6	7.10	−2.3	242.5	13.35
HMDS	1	7.4	0.35	21.0	6.37	−7.8	249.3	13.42
Toluene–cyclohexane	0.6–0.4	6.8	0.38	18.1	3.80	−3.9	253.4	13.87
HMDS–OMTS	0.7–0.3	5.0	0.18	27.4	6.70	−5.7	260.3	14.15

The electricity production and efficiency improvements, as a result of using mixtures instead of pure fluids, are smaller for the higher temperature heat source model than for the lower temperature heat source model.

5.3. Influence of the heat source temperature profile

Previously, only two sets of inlet and outlet temperatures (T_{in} , T_{out}) of the heat source were considered (See Table 2 for the values of a typical lower temperature heat source and Table 3 for a higher temperature heat source). Here, the effect of a variation of the heat source inlet temperature T_{in} , and of the temperature gradient $\Delta T_{src} = T_{in} - T_{out}$ over the heat exchanger is investigated. This temperature gradient is directly related to the thermal power recuperated from the heat source. The heat sink and cycle parameters are kept constant as in Table 2 and Table 3.

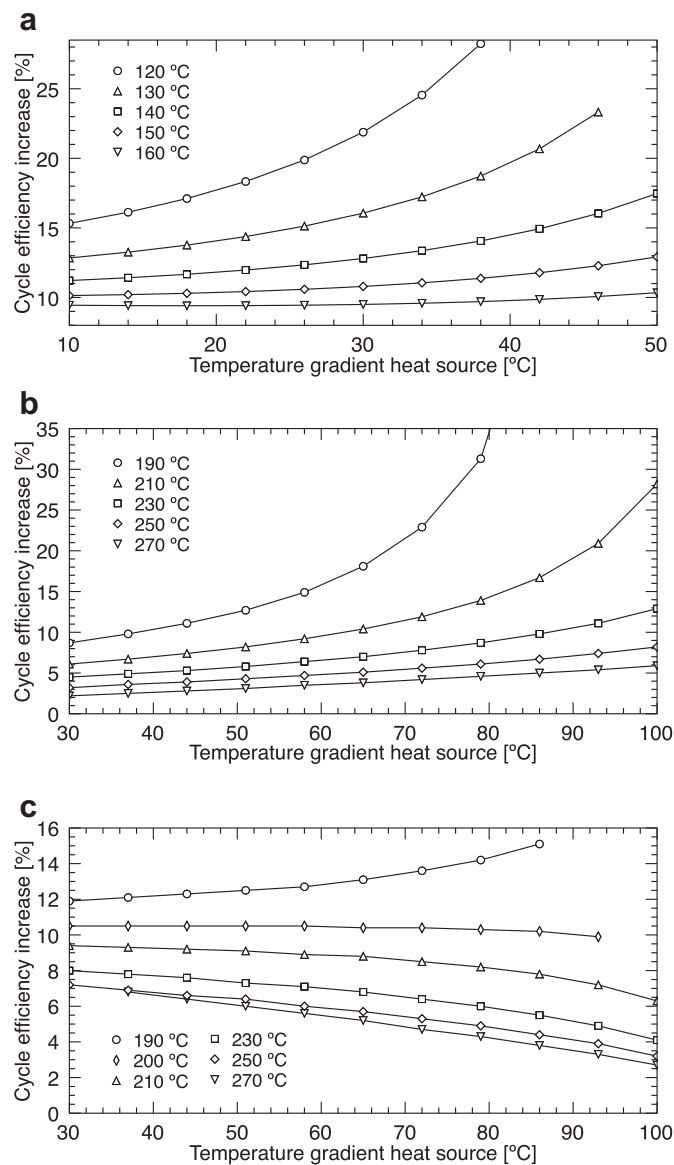


Fig. 6. Cycle efficiency increase as a function of the heat source temperature gradient over the evaporator for (a) 0.5/0.5 pentane/hexane compared to pure pentane, (b) 0.6/0.4 toluene/cyclohexane compared to pure toluene, and (c) 0.7/0.3 HMDS/OMTS compared to pure HMDS. Heat source temperatures are depicted in the legend. ORC with recuperator. Other parameters as in Table 2 (a) and Table 3 (b,c).

For the low temperature heat source, with a 0.5/0.5 mixture of pentane/hexane as working fluid, the cycle efficiency increase (with respect to pure pentane) as a function of ΔT_{src} is illustrated in Fig. 6(a), for several values of T_{in} . We note, first, that the efficiency increase is higher when the inlet temperature T_{in} of the heat source is lower. Secondly, the efficiency increase is also higher when the temperature drop ΔT_{src} over the heat exchanger is higher, i.e., when the outlet temperature T_{out} decreases. The same observations [Fig. 6 (b)] can be made for a toluene/cyclohexane mixture as working fluid in an ORC operating with the higher temperature heat source, but not for siloxane mixtures [Fig. 6 (c)]. Except for very dry fluids such as siloxanes, the benefit of using a mixture for heat sources with the considered parameters and inlet temperatures, increases when the temperature difference between the inlet and outlet of the heat exchanger is larger.

Bear in mind, however, that the cycle efficiency in absolute terms decreases with increasing ΔT_{src} , see Fig. 7 for a plot of the efficiency for both 0.5/0.5 pentane/hexane and pure pentane. The production of electricity, on the other hand, increases with ΔT_{src} until it reaches a maximum at ΔT_{src}^{opt} , and then decreases with ΔT_{src} . This is also illustrated in Fig. 7, again for 0.5/0.5 pentane/hexane and pure pentane. The generated electricity, rather than the overall cycle efficiency, will in most cases be the most relevant factor in the optimization of an ORC. The higher relative increase in electricity production of a mixture, compared with the corresponding pure fluid, for higher ΔT_{src} , makes mixtures as working fluid additionally promising, in particular for the ΔT_{src}^{opt} that yields maximum production.

For a heat source at 150 °C, the maximum net generated electricity versus the corresponding temperature gradient ΔT_{src}^{opt} for all considered pure fluids and mixtures is plotted in Fig. 8. Results with and without a recuperator are given. The mixture compositions are also optimal, see Table 6. The radius of a disk symbolizing a data point is proportional to the relative cycle efficiency. This figure clearly visualizes the increased electricity production potential of mixtures compared with pure fluids. The mixtures operate optimally at higher ΔT_{src}^{opt} compared to pure fluids, signifying an increase in recuperated power from the heat source. In addition, as already mentioned, the evaporation (condensation) process occurs at higher (lower) average temperature, resulting in a higher cycle efficiency. For the considered heat source and ORC, under optimal operation, an increase in electricity production of over 20% (without recuperator) and 23% (with recuperator) by using mixtures instead of pure fluids as medium is attainable. The results indicate that the inclusion of a recuperator, for this configuration,

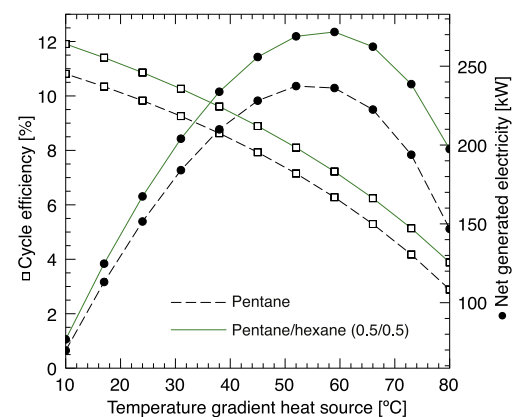


Fig. 7. Cycle efficiency (squares) and electricity production (bullets) of the ORC as a function of the heat source temperature gradient over the evaporator for a mixture of 0.5/0.5 pentane/hexane (solid lines) and pure pentane (dashed lines). The heat source inlet temperature is 150 °C. Other parameters as in Table 2.

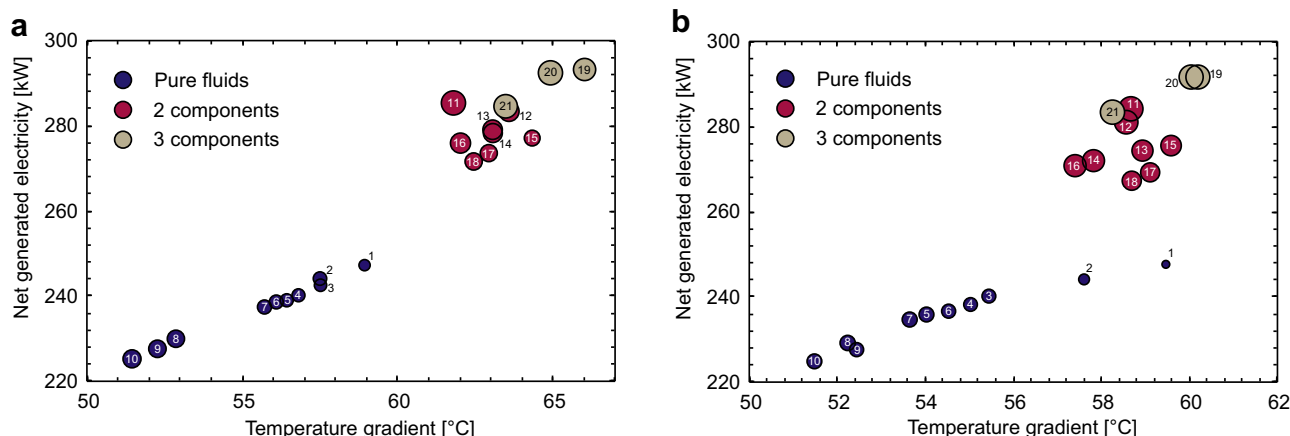


Fig. 8. Maximum net generated electricity versus corresponding optimal heat source temperature gradient for the fluids and mixtures given in Table 6, with optimal mixture compositions. (a) Without recuperator, (b) with recuperator. Disk radius is proportional to the relative cycle efficiency. The heat source is at 150 °C, other parameters as in Table 2.

actually decreases the electricity production slightly, and the extra cost and complexity of the system are not justifiable. In the remainder of the text, only results for a cycle without recuperator are mentioned. Again, 3-component mixtures only yield a minor increase in generated electricity compared to 2-component mixtures. In our survey, using R245fa/isopentane/isohexane (0.64/0.17/0.19) leads to the highest net electricity production (293.1 kW). We note that in current commercial applications, criteria other than thermodynamic performance can also be taken into account in the selection of a proper working fluid. For example, lower-efficiency fluids can be preferred over higher-efficiency fluids if the latter require oversized components due to a low vapor density.

Table 6

Simulation results for the ORC applied to the low temperature heat source for maximum achievable energy production. Listed are the medium numbers corresponding to Fig. 8, medium composition, optimal concentration, and net generated power, without and with a recuperator.

No.	Fluid	Without recuperator		With recuperator	
		C [mole frac.]	P_{net} [kW]	C [mole frac.]	P_{net} [kW]
1	Isobutane	1	247.2	1	247.6
2	R245fa	1	244.2	1	244.0
3	R365mfc	1	242.7	1	240.2
4	Isopentane	1	240.4	1	238.2
5	Isohexane	1	239.0	1	236.1
6	Pentane	1	238.6	1	236.6
7	Hexane	1	237.3	1	234.7
8	Cyclohexane	1	230.1	1	229.0
9	Cyclopentane	1	227.7	1	227.6
10	Toluene	1	225.3	1	225.0
11	Isopentane-cyclohexane	0.88–0.12	285.4	0.87–0.13	284.3
12	Isopentane-hexane	0.79–0.21	283.2	0.75–0.25	281.2
13	R245fa-pentane	0.32–0.68	279.2	0.32–0.68	274.5
14	Isopentane-isohexane	0.47–0.53	278.1	0.47–0.53	272.2
15	Isobutane-isopentane	0.23–0.77	277.1	0.26–0.74	275.7
16	Pentane-hexane	0.54–0.46	276.0	0.54–0.46	270.7
17	R245fa-R365mfc	0.40–0.60	273.5	0.41–0.59	269.2
18	R245fa-isopentane	0.27–0.73	271.7	0.28–0.72	267.5
19	R245fa-isopentane-isohexane	0.62–0.21–0.17	293.1	0.64–0.17–0.19	291.8
20	R245fa-pentane-hexane	0.60–0.32–0.08	292.6	0.61–0.30–0.09	291.5
21	Isopentane-isohexane-cyclohexane	0.80–0.10–0.09	284.6	0.80–0.09–0.11	283.2

6. Conclusions and final remarks

In summary, the use of suitable zeotropic mixtures as working fluids has a positive effect on the ORC performance in all investigated cases. An increase in cycle efficiency, for binary mixtures, of 15.7% and an increase in generated electricity of 12.3% is found possible for heat source and ORC cycle parameters for a lower temperature source (150 °C). The potential for efficiency and electricity production increase is less pronounced (6.0% and 5.5%, respectively) for a heat source type at a higher temperature (250 °C). In the considered cases, the addition of a third component to a binary mixture has only a small effect.

The potential increase in cycle efficiency and generated electricity is larger for lower temperature heat sources and when the temperature drop over the heat source exchanger is larger. When the ORC is optimized for operation with a low temperature heat source, the increase in electricity production (approximately 20%) by using mixtures is particularly remarkable.

The presented study was performed with an optimization tool that automatically selects the optimal pressures, mass flows and medium temperatures of the ORC under various assumptions. In reality, circumstances may prevent achieving these optimal settings. Also, the inner workings of the components (especially of the heat exchangers, but also of the pumps and turbine) and their optimization for mixtures as a medium, was not taken into account. The accuracy of the results can be further improved by considering the variation of the heat transfer coefficient of the mixture along the length of the heat exchangers. A number of publications have shown that the heat transfer coefficient of zeotropic refrigerant mixtures is smaller than linearly interpolated heat transfer coefficients based on those of pure constitutive refrigerants. See, Ref. [33].

Notwithstanding the potential beneficial effect on the efficiency and electricity production, the use of mixtures also has some disadvantages. Leakages present a larger problem than is the case for pure fluids, especially in the evaporation stage. Some mixtures are patented, barring their use or incurring an extra cost.

Nevertheless, the presented results demonstrate the potential for increased ORC performance by using mixtures, especially for low grade heat sources, if proper consideration is given to their selection. It is for the user to offset the increased performance against a higher installation complexity and cost and to incorporate the environmental impact of the fluids.

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Appendix A. Model equations

The ORC working fluid mass flow $\dot{m}$ is determined by the equality of the heat absorbed by the ORC working fluid and the heat given by the heat source medium in the evaporator:

$$\dot{Q}_{\text{evap}} = \dot{m}(h_{\text{evap}}^{\text{out}} - h_{\text{evap}}^{\text{in}}) = \dot{m}_{\text{src}}(h_{\text{evap,src}}^{\text{in}} - h_{\text{evap,src}}^{\text{out}}) \quad (\text{A.1})$$

where $h^{\text{in/out}}$ represents the enthalpy (mass unit) at the inlet/outlet of the evaporator and $\dot{m}_{\text{src}}$ is the heat source medium mass flow. We define the cycle efficiency as the ratio of the net generated power over the heat recovered from the heat source:

$$\eta_{\text{cycle}} = \frac{\dot{W}_{\text{gen}} - \dot{W}_{\text{pump}}}{\dot{Q}_{\text{evap}}} \quad (\text{A.2})$$

The generator power, or gross electric power, can be written as

$$\dot{W}_{\text{gen}} = \eta_{\text{gen}} \dot{W}_{\text{turb}} \quad (\text{A.3})$$

with η_{gen} the generator efficiency and the turbine power given by

$$\dot{W}_{\text{turb}} = \dot{m}(h_{\text{turb}}^{\text{in}} - h_{\text{turb}}^{\text{out}}) \quad (\text{A.4})$$

where $h_{\text{turb}}^{\text{in}}$ and $h_{\text{turb}}^{\text{out}}$ is the enthalpy of the working fluid at the inlet and outlet of the turbine. The latter is determined by

$$h_{\text{turb}}^{\text{out}} = h_{\text{turb}}^{\text{in}} - \eta_{\text{turb}}(h_{\text{turb}}^{\text{in}} - \bar{h}_{\text{turb}}^{\text{out}}) \quad (\text{A.5})$$

with η_{turb} the isentropic turbine efficiency and $\bar{h}_{\text{turb}}^{\text{out}}$ the enthalpy after the turbine if the expansion process occurred isentropically. Similarly, the pump power is given by

$$\dot{W}_{\text{pump}} = \dot{m}(h_{\text{pump}}^{\text{in}} - h_{\text{pump}}^{\text{out}}) \quad (\text{A.6})$$

with

$$h_{\text{pump}}^{\text{out}} = h_{\text{pump}}^{\text{in}} - \frac{h_{\text{pump}}^{\text{in}} - \bar{h}_{\text{pump}}^{\text{out}}}{\eta_{\text{pump}}} \quad (\text{A.7})$$

where η_{pump} is the isentropic pump efficiency and $\bar{h}_{\text{pump}}^{\text{out}}$ is the enthalpy after the pump if the process were isentropic. Conservation of energy in the recuperator implies that

$$h_{\text{evap}}^{\text{in}} - h_{\text{pump}}^{\text{out}} = h_{\text{turb}}^{\text{out}} - h_{\text{cond}}^{\text{in}} \quad (\text{A.8})$$

where $h_{\text{cond}}^{\text{in}}$ is the enthalpy at the inlet of the condenser. The set of equations is closed by the equality of the heat rejected by the ORC working fluid and the heat absorbed by the heat sink medium in the condenser:

$$\dot{m}(h_{\text{cond}}^{\text{in}} - h_{\text{cond}}^{\text{out}}) = \dot{m}_{\text{snk}}(h_{\text{cond,snk}}^{\text{out}} - h_{\text{cond,snk}}^{\text{in}}) \quad (\text{A.9})$$

As explained in the main text, the temperature differences over the evaporator and the condenser are determined by a classical pinch analysis [32]. The recuperator effectiveness is set by the temperature difference

$$\Delta T_{\text{recup}} = T_{\text{cond}}^{\text{in}} - T_{\text{pump}}^{\text{out}} \quad (\text{A.10})$$

between the inlet of the condenser and the outlet of the pump. Thermodynamic state functions are calculated with REFPROP [29].

The mixing rules for mixtures are set in REFPROP. It would lead us too far here to list all the data, which can be retrieved within the package, including references to the scientific literature. For hydrocarbons, for example, the Kunz and Wagner model [34] is applied. In Table A.1 we present values for the model parameters used in this study.

Table A.1

Parameters in the Kunz and Wagner model for mixtures of hydrocarbons.

Fluid	Conc	BetaT	Gamma	BetaV	Gamma	Fij
Pentane-hexane	0.50–0.50	1.0	1.008	1.0	1.0025	0.0
Isopentane-cyclohexane	0.79–0.21	1.0	1.0	1.0	0.9961	0.0
Isopentane-hexane	0.61–0.39	1.0	1.0012	1.0	1.003	0.0
Toluene-cyclohexane	0.6–0.4	1.0	0.9979	1.0	0.9979	0.5865

Appendix B. Error analysis

The model was validated by comparisons with the Cycle-Tempo [35] energy system simulation application. For pure fluids, the uncertainties in the used thermodynamic state functions given by REFPROP [29] are less than 2% in the considered regimes. We compared values given by REFPROP and the Peng–Robinson equation of state [36], see Table B.1 for the difference in results for the temperature state function for several states of pentane, hexane, and 0.5/0.5 pentane/hexane. Optimizations were performed with the generalized reduced gradient method [37] with a convergence tolerance fixed at 10^{-4} . Multidimensional functions were plotted to ascertain that the calculated maxima are global.

Table B.1

Difference [%] between REFPROP and the Peng–Robinson equation of state for the temperature state function at given pressure, vapor quality states.

Pressure [bar]	1		10	
	0	1	0	1
Vapor quality				
Pentane	0.65	0.65	0.04	0.04
Hexane	0.21	0.21	0.05	0.05
Pentane-hexane (0.5–0.5)	1.14	0.76	0.22	0.21

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